

CHEYENNE, WY, USA

WMO: 725640

Lat: 41.158N

Lon: 104.807W

Elev: 6113

StdP: 11.73

Time Zone: -7.00 (NAM)

Period: 94-19

WBAN: 24018

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-3.1	3.1	-10.9	3.8	4.0	-5.8	5.0	13.0	40.0	36.9	36.2	36.2	9.1	0	0.709

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	26.5	89.8	58.1	87.3	57.6	84.5	57.3	62.9	77.6	61.7	77.0	60.5	75.9	11.6	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
58.8	93.2	66.1	57.1	87.6	65.4	55.5	82.3	64.7	31.4	77.4	30.4	77.2	29.5	75.7	69.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
33.3	28.5	25.3	DB	-12.5	94.3	6.3	2.3	-17.1	95.9	-20.8	97.3	-24.4	98.5	-29.0	100.2	(4)
			WB	-13.1	65.9	6.2	1.6	-17.5	67.1	-21.1	68.0	-24.6	68.9	-29.1	70.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	47.0	29.4	29.3	37.0	42.6	51.7	63.0	70.1	68.0	59.6	46.5	36.6	28.6
	DBStd	17.38	10.89	11.37	10.30	9.37	8.68	6.98	5.01	5.05	8.27	9.77	10.96	11.28
	HDD50	3250	638	582	414	251	86	4	0	0	21	178	413	663
	HDD65	6972	1102	1001	869	672	416	116	16	30	194	575	853	1128
	CDD50	2142	1	1	9	29	139	393	624	557	310	69	10	0
	CDD65	387	0	0	0	0	3	55	174	123	32	0	0	0
	CDH74	5344	0	0	0	4	118	927	2151	1508	605	31	0	0
	CDH80	1830	0	0	0	0	18	303	862	492	154	1	0	0
Wind	WSAvg	12.0	13.9	13.5	13.3	13.2	11.7	10.7	9.7	9.6	10.1	11.7	12.7	13.5
Precipitation	PrecAvg	14.5	0.4	0.4	1.0	1.5	2.3	2.2	2.0	1.7	1.4	0.8	0.6	0.4
	PrecMax	20.7	2.7	1.3	3.7	5.0	6.0	4.6	5.0	6.7	4.5	2.0	2.5	1.5
	PrecMin	5.9	0.0	0.0	0.1	0.4	0.1	0.1	0.4	0.2	0.0	0.0	0.0	0.0
	PrecStd	3.4	0.4	0.3	0.7	1.0	1.3	1.1	1.1	1.1	1.1	0.5	0.6	0.3
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	58.0	61.1	69.2	73.6	82.4	91.3	93.6	91.0	88.3	77.7	68.5	58.3
		MCWB	39.2	41.3	45.0	47.5	53.7	56.9	58.7	58.2	56.9	51.1	45.1	39.7
	2%	DB	52.5	55.3	63.9	69.6	77.2	87.1	90.6	88.0	84.2	73.4	63.0	52.3
		MCWB	37.1	38.5	42.7	46.5	52.4	56.5	58.6	57.5	55.3	49.3	42.9	37.1
	5%	DB	48.5	50.2	59.4	65.5	73.3	83.5	88.2	85.5	81.0	69.4	58.4	48.5
		MCWB	35.4	36.5	40.8	45.0	51.5	56.0	58.3	57.4	54.1	47.8	41.0	35.7
	10%	DB	45.0	45.9	54.6	60.8	68.8	79.6	85.2	82.6	77.2	64.1	53.5	44.8
		MCWB	33.7	34.2	39.0	43.3	50.2	56.1	58.4	57.4	53.2	46.0	38.8	33.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	40.8	42.0	46.1	50.3	58.4	63.2	65.2	64.0	61.5	53.4	46.4	40.4
		MCDB	54.5	58.9	65.0	67.8	73.3	77.2	78.9	78.2	78.9	71.0	65.0	55.6
	2%	WB	38.1	39.3	43.8	47.9	55.7	61.1	63.5	62.2	59.0	51.2	43.8	38.1
		MCDB	50.4	53.3	61.1	64.0	69.4	75.9	77.9	77.6	74.7	68.3	61.5	50.8
	5%	WB	36.1	37.0	41.7	46.0	53.5	59.4	62.2	61.1	57.1	49.2	41.6	36.1
		MCDB	47.6	49.3	57.6	62.1	67.5	74.5	77.1	76.8	71.6	64.6	57.0	47.7
	10%	WB	34.0	34.7	39.4	44.2	51.7	57.8	61.0	59.8	55.3	47.1	39.3	34.0
		MCDB	44.4	45.1	53.3	59.3	65.4	73.4	76.3	74.8	70.2	62.2	52.3	44.3
Mean Daily Temperature Range	5% DB	MDBR	20.1	20.6	22.7	22.4	22.6	25.8	26.5	26.2	25.9	23.5	21.5	19.6
		MCDBR	23.7	25.5	29.8	31.6	30.1	32.3	31.2	31.0	31.9	31.9	28.5	23.7
		MCWBR	13.2	14.1	14.4	14.3	12.1	11.6	9.8	9.7	11.4	14.1	14.2	13.4
	5% WB	MCDBR	22.3	24.5	28.2	28.7	26.6	27.1	25.7	25.9	26.4	28.5	27.1	22.7
		MCWBR	13.1	14.0	14.1	13.9	12.0	12.0	9.8	9.6	10.6	13.6	14.1	13.3
Clear-Sky Solar Irradiance	taub	0.233	0.240	0.262	0.287	0.308	0.315	0.341	0.336	0.298	0.262	0.245	0.235	
	taud	2.600	2.587	2.536	2.492	2.489	2.492	2.454	2.455	2.553	2.630	2.581	2.568	
	Ebn at Noon	305	317	318	313	306	303	294	293	300	304	296	295	
	Edh at Noon	21	24	29	33	34	34	35	34	29	23	21	20	
All-Sky Solar Radiation	RadAvg	744	1044	1436	1710	1925	2274	2188	1922	1610	1134	806	627	
	RadStd	30	61	103	113	164	116	86	55	91	84	46	37	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (21 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page